*f ∘ *g = *gf
f* ∘ g* = fg*
f* ∘ *g = *g ∘ f*

TODO
- comma category

Num category

x	object
x+n: x → x+n	arrow
x+n ∘ (x+n)+m: x → x+n+m	composition

Functor
*3: Num → Num

*3 x = 3x	object
*3 (x+n): 3x → 3(x+n)	arrow

*3 (x+n ∘ (x+n)+m): 3x → 3(x+n+m)
*3 (x+n) ∘ 3((x+n)+m) = *3 (x+n+m)

Adjoint
x + *2 *3 x: x → x + 6x
x' + (x - x'): x' → x
*6(x' + (x - x')): 6x' → 6x

(x + *2 *3 x) ∘ (x' + (x - x')): x' → x' + 6x
*6(x' + (x - x')) ∘ (x + *2 *3 x):

Functor
Let A, B be categories

!A = A^op

Functor

F: A → B	
FX ∈ B	**object map** for X ∈ A
Ff: FX → FX'	**arrow map** for f: X → X' ∈ A

Contravariant functor

F: !A → B	
FX ∈ B	**object** map for X ∈ A
Ff: FX' → FX	**contravariant arrow** map for f:X→X' ∈ A

==========

----- Composition/base map -----

names:
base map
base point
object map
linear map
conjugate
convolution
local ...
projection

homomorphism

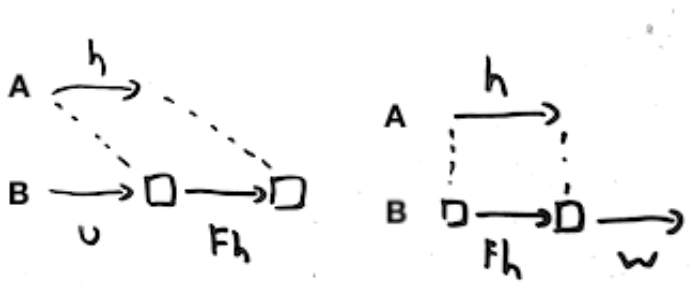
Hom functor
Let F, G: A→B

f*: {_→Y} ⇒ {_→Y'} **arrow map** for f:Y→Y'
f*(h) = f ∘ h
σ*: {_→FX} ⇒ {_→GX} **arrow map** for σ: F ⇒ G
σ*(h) = σX ∘ h
*g: {Y→_} ⇒ {Y'→_} **contravariant map** in g: Y'→Y
*g(h) = h ∘ g
*τ: {FX→_} ⇒ {GX→_} **contravariant map** in τ: G⇒F
*τ(h) = h ∘ τ,X

Base map (aka offset)
Let
F: A → B
u: Y → FX, w: FX → Y

Base
F * u: {X→_} ⇒ {Y→F_} map by base (offset) u: Y → FX
F * u(h) = Fh ∘ u

Co-base (contravariant)
w * F: {_→X} ⇒ {F_→Y} map by co-base w: FX → Y
w * F(h) = w ∘ Fh



Base map set

Sets
Y→F[X→_] = { F * u | u: Y → FX }
F[_→X]→Y = { F * w | w: FX→Y }

Base bifunctor ·→∗ [X→_]
Base ·→∗ [X→_]: B x B^A → Set **object map** for (Y, F) ∈ !B x B^A
Base Y→F[X→_] = { F * u | u: Y → FX }

σ*: [Y→F[X→_]] → [Y→G[X→_]] **arrow** map in σ: F ⇒ G ∈ B^A
σ* (F * u) = G * (σX ∘ u)

*f: [Y→F[X→_]] → [Y'→F[X→_]] **contra arrow map** in f: Y→Y' ∈ B

*f (F * u) = *f ∘ (F * u)

Base map lattice

Let

K: J→A L: J→B

F: A→B

Base map lattice

F * ηX ∈ L ⇒ F[K→_] for base η: L⇒FK

F * ηX ∈ LX→F[KX→_]

F * η : {L・→_} ⇒ {K・→F_}

where for any f: X→X'

FKf ∘ ηX = ηX' ∘ Lf

Co-base

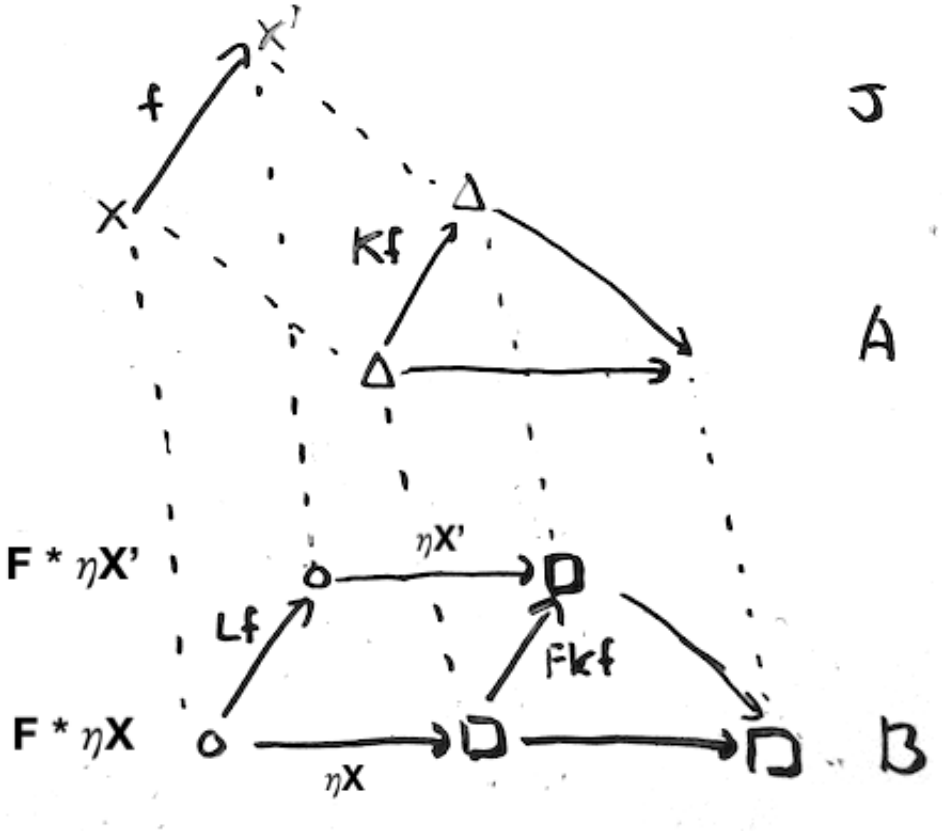
ε * F ∈ F[_→K]⇒L for co-base ε: FK⇒L

εX * F ∈ F[_→KX]→LX

ε * F: {_→K・}⇒ {F_→L・}

where for any f: X→X'

Lf ∘ αX = αX' ∘ FKf

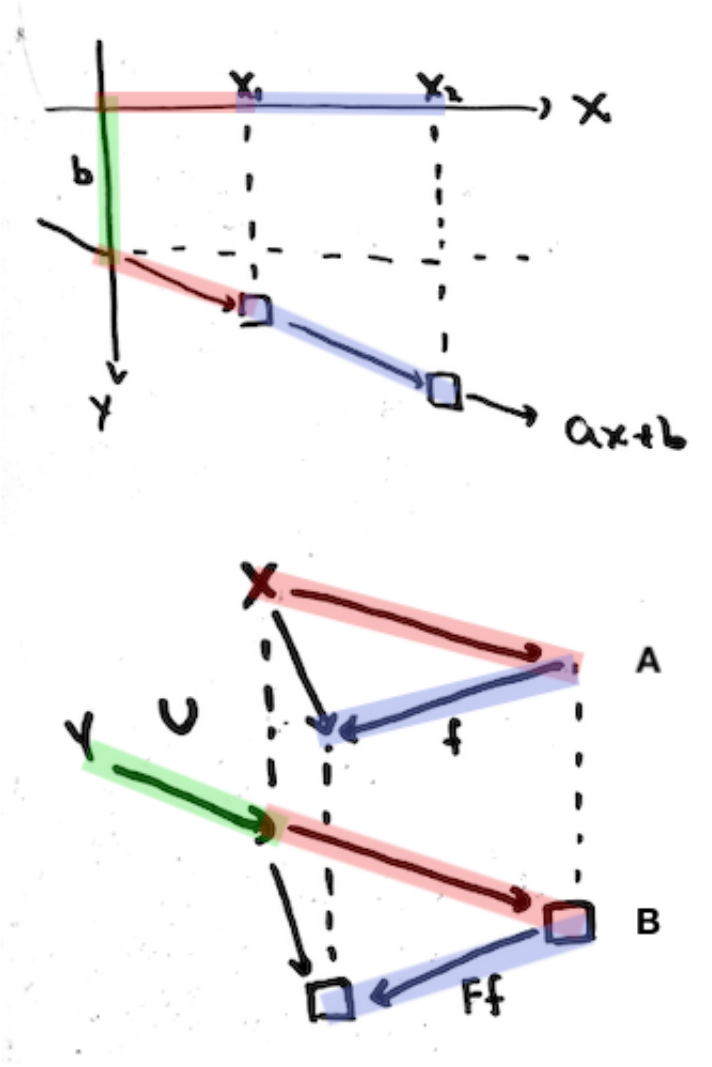


Note: Base map ~ Linear map

Base map corresponds to a linear map for categories

Line		**Base map**	
offset	b	offset	u: Y → FX
factor	a	factor	F: A → B

equation	f(x) = ax + b	map	F * u (h)	= Fh ∘ u
linearity	f(x' + x) = ax' + f(x)	linearity	F * u (h' ∘ h)	= Fh' ∘ (F * u) (h)



*** Special cases**

- **Composition** aka Yoneda embedding

F = Identity IA

Composition

*u = I * u: {X→_} ⇒ {Y→_}

w* = w * I: {_→Y} ⇒ {_→X}

Hom functor (**Yoneda embedding**)

Hom: A^op → Set^A

Hom X = {X→ _} object

Hom u^op: {X→_} ⇒ {Y→_} composition arrow

Hom u^op = *u^op

for u: Y → X

- **Element** aka application

Special case where F = point set {*→F_}

Let

F: A → Set

G: A^op → Set^op i.e. contravariant

Base element x $\in$ FX
F * x: {X→_} ⇒ {* → F_} ≅ F
F * x(h) = Fh(x)

Co-base element x $\in$ Gx (contravariant)
x * G: {_→X} ⇒ G contravariant map
x * G(h) = Gh(x)
note: Gh:GX→GX' if h:X→X'

- **0-functor** 0 → A

0: 0 → A
0* $\in$ Y → 0[-]
0* - no arrows

*0 $\in$ 0[-] → Y
*0 - no arrows

- **Isomorphism** aka universal property
Let
F: A → B
Y $\in$ B, GY $\in$ A

Universal property
F * u: {GY→_} ⇒ {Y→F_} **isomorphic map**
u: Y→FGY universal arrow

Co-universal property w * F
w * F: {_→GY} ⇒ {F_→Y} **isomorphic contravariant** map
w: FGY→Y couniversal arrow

- **Adjoint**
Universal + natural in both X, Y

Adjoint
φ: {FX→_} ⇒ {X→G_} **universal**
φ = G * ηX
ηX: X → GFX base
ηX' ∘ f = FGf ∘ ηX with f:X→X' **natural in X**

-φ: {_→GY} ⇒ {F_→Y} **couniversal**
-φ = εX * F
εY: FGY → Y co-base
g ∘ εY = εY' ∘ GFg with g:Y'→Y **co-natural in Y**

*** Basic properties**
Proposition: identities
- Composition
Base cases:

Base
*u ∘ *v = *vu

$u^* \circ v^* = uv^*$
$u^* \circ {}^*v = {}^*v \circ u^*$

Functor
${}^*F = {}^*F$
$F^* \circ G^* = {}^*F \circ {}^*G = FG^*$
${}^*u \circ F^* = F * u$
$u^* \circ F^* = u * F$
$F^* \circ {}^*u = F * Fu = {}^*Fu \circ F^*$
$F^* \circ u^* = Fu * F = Fu^* \circ F^*$

Hence:

$${}^*u \circ (F * v) = {}^*u \circ {}^*v \circ F^* = {}^*uv \circ F^* = F * vu$$
$${}^*u \circ (v * F) = {}^*u \circ v^* \circ F^* = v^* \circ {}^*u \circ F^* = v^* \circ (F * u)$$

$$u^* \circ (v * F) = u^* \circ v^* \circ F^* = uv * F^* = uv * F$$
$$u^* \circ (F * v) = u^* \circ {}^*v \circ F^* = {}^*v \circ u^* \circ F^* = {}^*v \circ (u * F)$$

$$(F * u) \circ {}^*v = {}^*u \circ F^* \circ {}^*v = {}^*u \circ (F * Fv) = F * (Fv \circ u)$$
$$(F * u) \circ v^* = {}^*u \circ F^* \circ v^* = {}^*u \circ Fv^* \circ F^* = Fv^* \circ {}^*u \circ F^* = Fv^* \circ (F * u)$$

$$(u * F) \circ v^* = u^* \circ F^* \circ v^* = u^* \circ (F * Fv) = (u \circ Fv) * F$$
$$(u * F) \circ {}^*v = u^* \circ F^* \circ {}^*v = u^* \circ {}^*Fv \circ F^* = {}^*Fv \circ u^* \circ F^* = {}^*Fv \circ (u * F)$$

General:

$$(F * u) \circ (G * v) = {}^*u \circ (FG * Fv) = FG * (Fv \circ u)$$
$$(u * F) \circ (v * G) = u^* \circ (Fv * FG) = (u \circ Fv) * FG$$

$$(F * u) \circ (v * G) = {}^*u \circ (Fv * FG) = Fv^* \circ (FG * u)$$
$$(u * F) \circ (G * v) = u^* \circ (FG * Fv) = {}^*Fv \circ (u * FG)$$

- **Horizontal**

$\sigma^* \circ (F * u) \quad = \quad G * (\sigma X \circ u) \qquad$ for $\sigma: F \Rightarrow G$

==========
proof

- Horizontal
For any g: X → X'

G * (σX ∘ u) g	=	Gg ∘ σX ∘ u	
	=	σX' ∘ Fg ∘ u	naturality of σ
	=	σ ∘ (F * u) g	

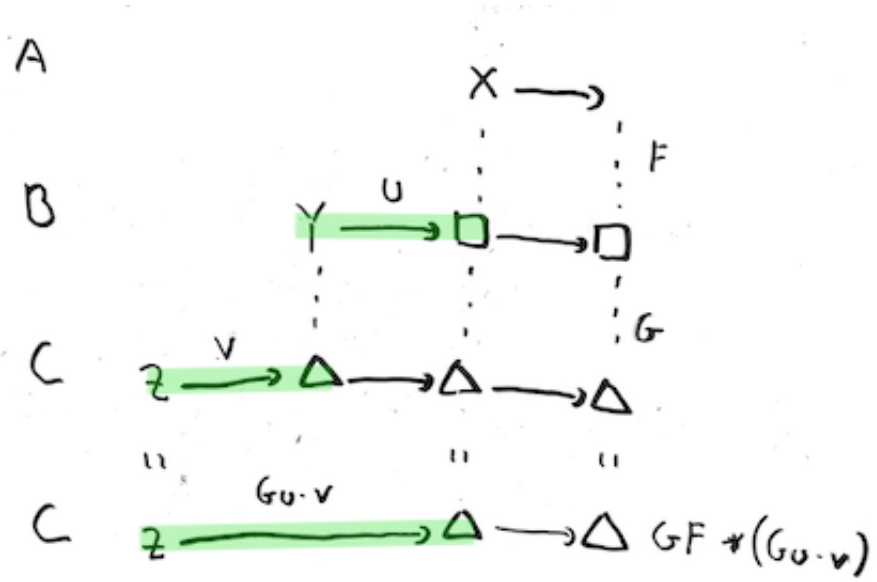
QED
==========

Corollary: Closure
Base maps are **closed** under composition
i.e. let
F: A → B G:B → C
F * u ∈ Y→F[X→_]
G * v ∈ Z→ G[Y→_]

Then composition is also a base map:

(G * v) ∘ (F * u) = GF * (Gu ∘ v) ∈ Z → GF[X→_]

GF: A → C	functor
Gu ∘ v: Z → GFX	base



Theorem: Fundamental property (Yoneda)
Let F: A → B
There is a 1-1 correspondence

base map		**natural transformation**
$F * u \in Y \to F[X \to _]$	<=>	φ: $\{X \to _\} \Rightarrow \{Y \to F_\}$

where

$\varphi = F * u$ where $u = \varphi 1X$: $Y \to FX$

special case: If F = IA, then φ is just **composition (aka Yoneda embedding)**

$\varphi = *u$ where $u: Y \to X \in FX$

==========

Proof:

- natural => base map

Let φ natural. For any $h: X \to Y$

$\varphi h = \varphi(h \circ 1X)$

$= Fh \circ \varphi 1X$ naturality

$= F * u\ (h)$ $u = \varphi 1X$

- base map => natural

Let $\varphi = F * u$. Then

$\varphi \circ f^* = (F * u) \circ f^*$

$= Ff^* \circ (F * u)$

$= Ff^* \circ \varphi$

=> natural

QED

==========

Corollary

Hom is **full** and **faithful**

i.e. every Hom u^op: $\{X \to _\} \Rightarrow \{Y \to _\}$ is of the unique form $\varphi = *u$

Corollary: isomorphism

If F * u and G * v are isomorphic, then so is the composite GF * (Gu ∘ v)

Proof: composition of isomorphisms G * v and F * u is an isomorphism QED

Proposition: Embedding

there exists $f: Y \to X$ isomorphism <=> there exists φ: $\{X \to _\} \Rightarrow \{Y \to _\}$ a natural isomorphism

φ **= *f** i.e. each φX is isomorphic

Hence

$X \cong Y$ <=> there exists natural isomorphism φ: $\{X \to _\} \Rightarrow \{Y \to _\}$

==========

Proof

f isomorphism	=>	φ = *f natural isomorphism
φ natural isomorphism	=>	if φ = *f, then φX = f is isomorphic

QED

==========

<u>* Vertical transformation</u>

Vertical transformation: A → B → B'

Composition of base maps.

i.e. Let

F:A→B	F * u ∈ Y→F[X→_]
F':A'→B'	F' * u' ∈ Y'→F'[X'→_]
K': A → A'	K' * v' ∈ X'→K'[X→_]
K: B → B'	K * v ∈ Y'→K[Y→_]

If

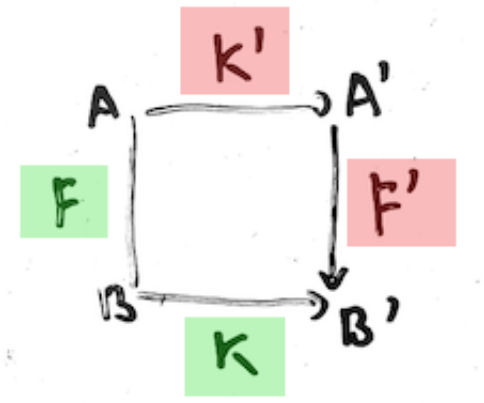
F'K' = KF: A → B'

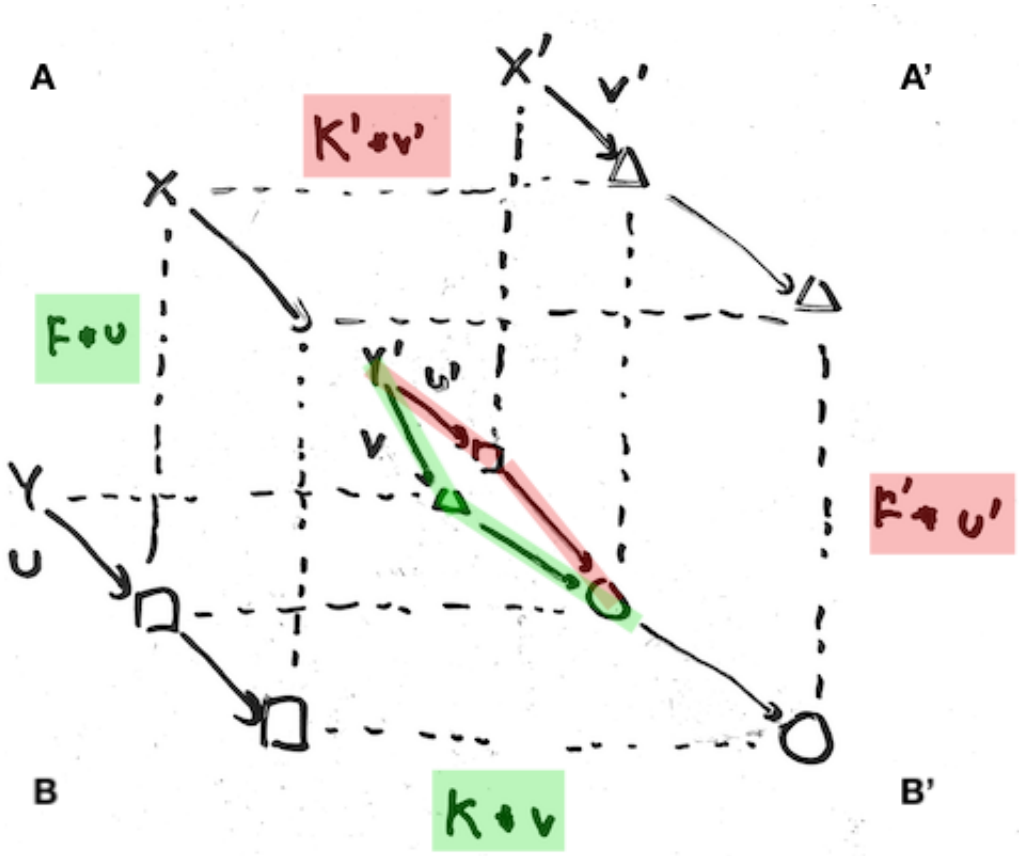
(F' * u') ∘ (K' * v') = (K * v) ∘ (F * u)

i.e. base map F'K' * (F'v' ∘ u') = KF * (Ku ∘ v) ∈ Y'→KF[X→_]

then **(K * v, K' * v')** is a vertical transformation of F * u to F' * u'.

Or dually, (F * u, F' * u') is a vertical transformation of K' * v' to K * v





Special cases

unit: i.e. F or F = identity
*u' ∘ (F * u) = F * (u' ∘ u)
(F' * u'') ∘ *u = F' * (u ∘ u')

functor map aka vertical transformation: i.e. u or u' = 1
F'* ∘ (F * u) = F'F * F'u
(F' * u') ∘ F* = F'F * u

Theorem: base
Vertical transformation is determined by the bases
i.e. if F'K' = FK, then

(F' * u') ∘ (K' * v') = (K * v) ∘ (F * u) <=> **F'v' ∘ u' = Ku ∘ v**

==========
Proof: just by the identity F'K' * (F'v' ∘ u') = KF * (Ku ∘ v)
=>) let h = 1. i.e.
(F'v' ∘ u') = F'K' * (F'v' ∘ u') (1) = KF * (Ku ∘ v) (1) = (Ku ∘ v)
<=) trivial
QED
==========

<u>* Horizontal transformation</u>
Horizontal transformation F ⇒ F'

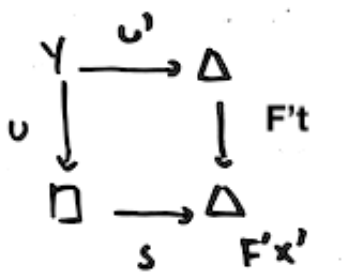
Let
F, F':A→B

u: Y→FX u': Y→F'X'

If there exists (s: FX→F'X, t:X'→X) such that

s ∘ u = F't ∘ u' : Y→F'X'

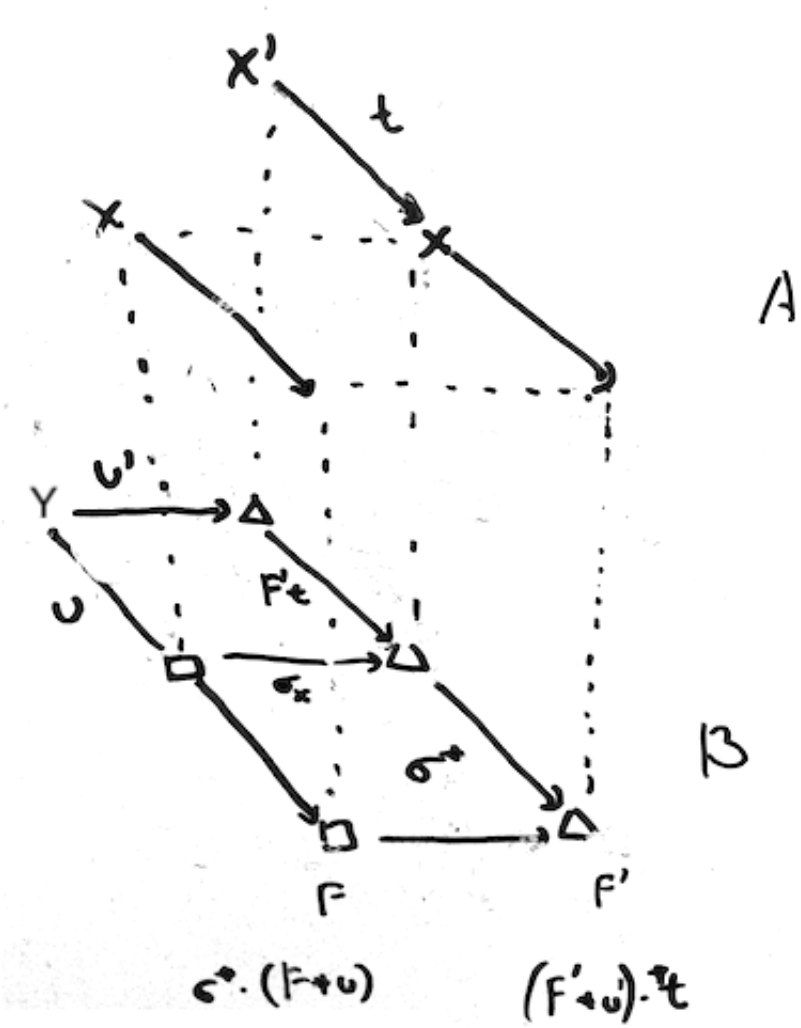
then (s, t) is a horizontal transformation of u to u'.



Proposition: Horizontal transformation
Let
F, F': A→B
F * u ∈ Y→F[X→_] F' * u' ∈ Y→F'[X'→_]
σ: F ⇒ F'

Then

base map		**base**
σ* ∘ **(F * u) = (F' * u') ∘ *t** ∈ Y→F'[X→_]	<=>	σ**X ∘ u = F't ∘ u'** for all X i.e. (σX, t) is a horizontal transformation
	<=>	F' * (σX ∘ u) = F' * (F't ∘ u')



==========

Proof
Use naturality

F' * (σX ∘ u) (h)	=	F'h ∘ σX ∘ u	
	=	σX' ∘ Fh ∘ u	by naturality
	=	σX' ∘ (F * u) (h)	
F' * (F't ∘ u')	=	(F' * u') ∘ *t	
=> σ* ∘ (F * u)	=	F' ∘ (σX ∘ u)	
	=	F' ∘ (F't ∘ u')	by definition
	=	(F' * u') ∘ *t	

QED

==========

Corollary: Horizontal transformation
Let
G, G': B→A F, F': A→B
G * $\eta \in * \Rightarrow G[F* \rightarrow _]$ G' * $\eta' \in * \Rightarrow G'[F'* \rightarrow _]$
σ: G ⇒ G' τ: F' ⇒ F

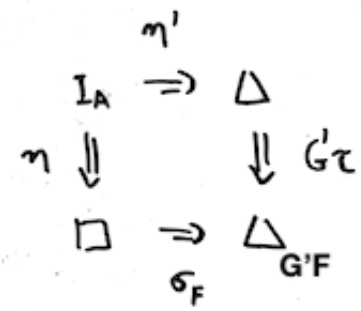
Then

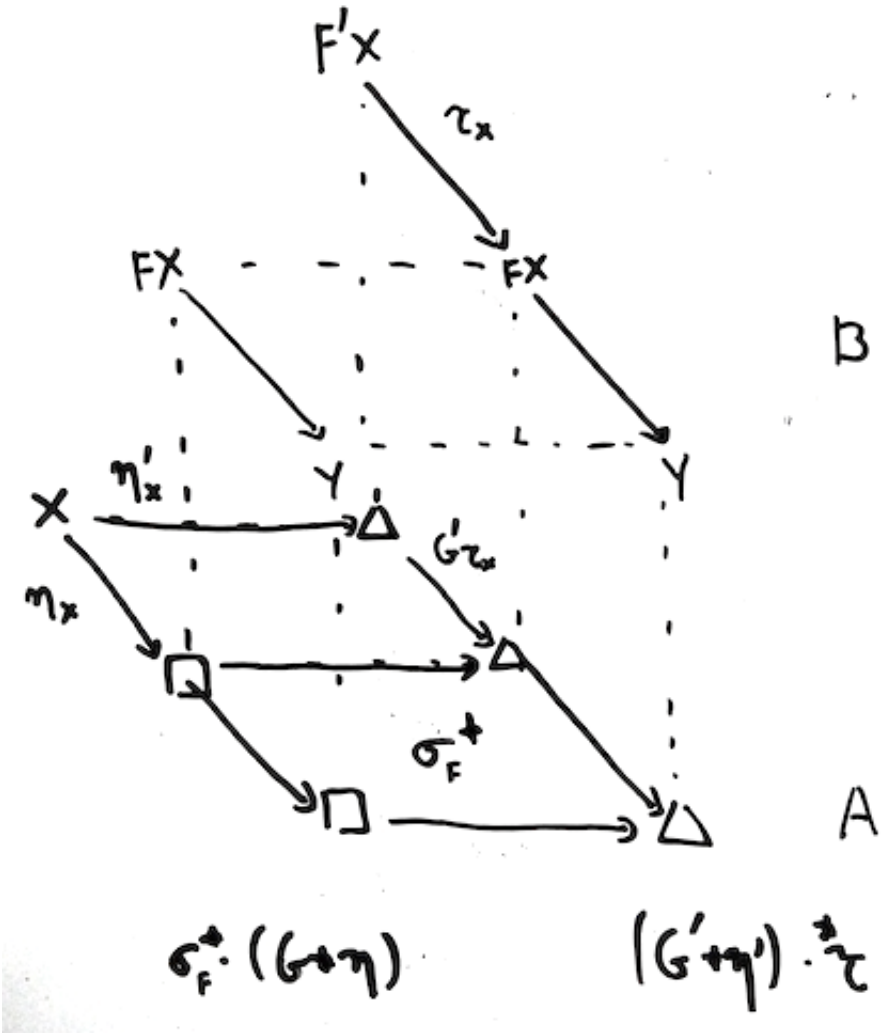
base map	**base**

$\sigma F^* \circ (G * \eta) = (G' * \eta') \circ {}^*\tau$ <=> $\sigma \mathbf{F} \circ \eta = \mathbf{G'}\tau \circ \eta'$

$\in * \Rightarrow G'[F* \to _]$ $(\sigma F, \tau)$ is a horizontal map

<=> $G' * (\sigma F \circ \eta) = G' * (G'\tau \circ \eta')$





$\sigma_F^+ \cdot (G * \eta)$ $(G' * \eta') \cdot {}^*\tau$

==========

----- Isomorphic base maps aka universal property----

Universal property
A universal property is just a bijective base map

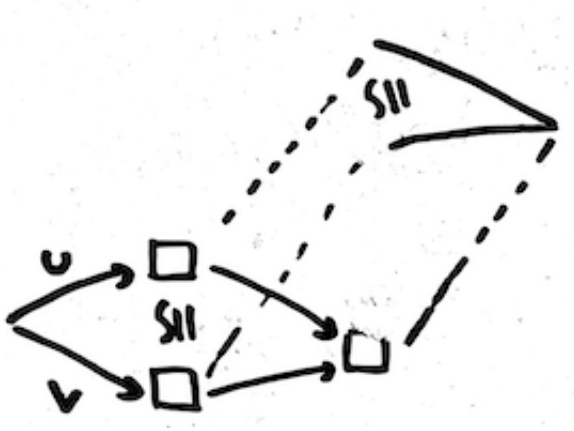
Proposition: uniqueness
The base is unique up to isomorphism

i.e.

both isomorphic
F * u ∈ Y → F[X→_]
F * v ∈ Y→F[X'→_] <=> **X ≅ X'**

In this case

v = Ff ∘ u <=> f: X→X' isomorphism
= F * u (f) **-(F * u) v = f**



==========
Proof:
=> Suppose F * v is isomorphic

v = F * u (v') for some unique v': X → X'
u = F * v (u') for some unique u': X' → X

=>
u = F * v (u')
= F * (F * u (v')) (u')
= F(u'v') ∘ u
= F * u (u'v')

but also

u = F * u (1)

=> u'v' = 1X

Similarly, v'u' = 1X'
Also

v= F* u (v') = Fv' ∘ u

<= If f: X→X' is an isomorphism

F * (Ff ∘ u) = (F * u) ∘ *f
=> isomorphic since both f and F*u are isomorphic

QED
==========

Proposition: preservation
Let F * u be isomorphic

F * u **preserves** isomorphisms <=> u: Y→ FX isomorphic
i.e. if f is isomorphic then so is F * u (f)

==========

Proof:
=>) Suppose F * u preserves isomorphism
Then u = F * u(1) is isomorphic

<=) Suppose u is isomorphic.

f isomorphic
=> Ff isomorphic — functors preserve isomorphism
=> F * u(f) = Ff ∘ u isomorphic — compositions preserve isomorphism

QED
==========

Proposition: functor composition
Let
F: A → B, I: C → A, J: B → D
F * u ∈ Y→F[X→_]

Then

I: C → A isomorphic => FI * u ∈ Y→FI[I^-1(X) → _] is **isomorphic**
FI * u = (F * u) ∘ I*

J: B → D isomorphic => JF * Ju: {X → _} → {JY → JF_} is **isomorphic**
JF * Ju = J(F * u)

==========

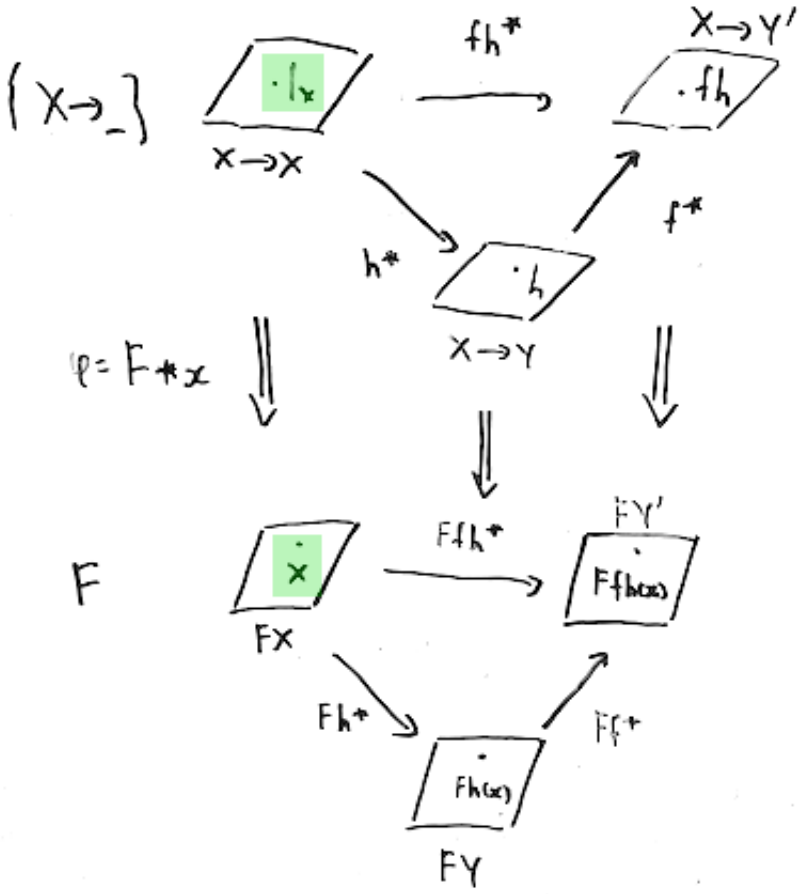
----- Representation ----

Representation (of arrow)
Representable
Representing **base arrows { X → Y }** in some other form **FY**
Let F: A → Set.
If for some X ∈ A

φ: { X →_ } ⇒ F

is isomorphic, then φ is a **representation as F**.
F is **representable** if such a φ exists.
Correspondence

	{ X → _ }	**F**
object	hom set { X → Y } arrow X → Y	set FY element y ∈ FY
arrow	composition f*: { X→Y } → { X→Y' } f* (h) = f ∘ h	function Ff: FY → FY'



Theorem: fundamental theorem of representation
A representation is precisely a **universal property**
i.e. Let natural φ: { X →_ } ⇒ F. Then by the Yoneda Lemma,

φ **= F * x** where x = φ(1X)

Hence

φ is a **representation** <=> φ = F * x is a **universal property**

Thus, in particular, a representation is determined by the **single** universal element x = φ(1X).

==========
proof: Yoneda lemma
==========

Hence, the collection of all base maps F * x is represents all possible representations

Corollary: category of representations
The objects of the comma category (Hom ↓ 1F) represent all possible potential representations
Each **terminal** object is precisely a representation.
i.e. in comma category (Hom ↓ 1F)

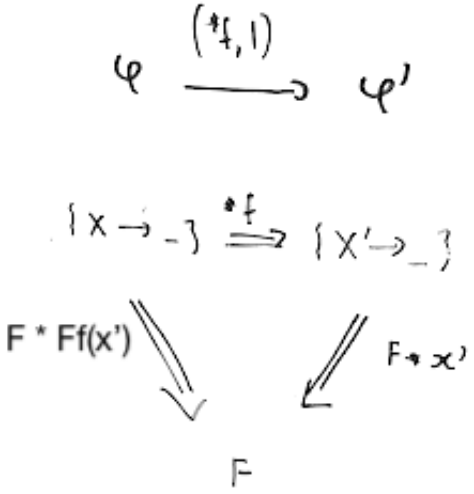
object
φ: { X→_ } ⇒ F base map F * x, for any X ∈ A, x ∈ FX
φ = F * x

arrow
(*f, 1): $\varphi \to \varphi'$ for any f: X'→X
iff $\varphi = \varphi' \circ$ *f *f: { X→_ } ⇒ { X'→_ }

Then:

- **arrow**: each arrow (*f, 1): $\varphi \to \varphi'$ is of the form

(*f, 1): F * Ff(x') → F * x' for x' ∈ FX'
F * Ff(x') = (F * x') ∘ *f



- **representation**:

φ = F * x is a representation <=> F * x is a **terminal object**

==========
Proof
- arrow (*f, 1): $\varphi \to \varphi'$
If φ' = F * x' for some x' ∈ FX'
φ = $\varphi' \circ$ *f
= (F * x') ∘ *f
= F * Ff(x')

- terminal
Idea: by Yoneda, there is 1-1 correspondence between a transformation φ = F * x and x ∈ FX
i.e. can treat φ and x as **equivalent**

By definition, for a fixed X, a representation if a 1-1 correspondence between any f: X → Y <=> y ∈ FY

Ff(x) = y
f: X → Y <=> y ∈ FY
<=> F * y for y ∈ FY 1-1 correspondence by Yoneda
<=> (*f, 1): F * y → F * x for y ∈ FY

=> i.e. for any base map F * y, there is a unique f: X → Y
(*f, 1): F * y → F * x

Hence terminal
QED
==========

Proposition: continuity (limit)
Let F: A → Set. If F has a representation, then F is **continuous**.
In particular, if F is not continuous, then F is not representable.

==========
proof: hom-functor is continuous so any F isomorphic to it must be continuous.
==========

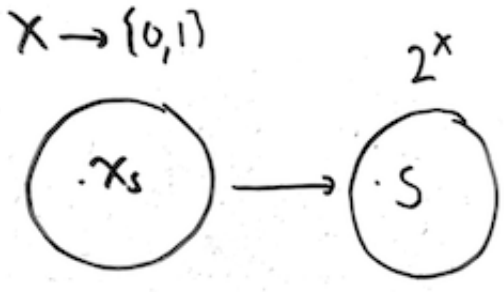
*** Examples**
- **Universal property** F * u ∈ Y→F[X→_]
Let F: A → B.
A universal property is a representation of **arrows X → _ in A** as arrows **Y → F_ in B**

Representation

F * u: { X → _ } ⇒ { Y → F_ }
F * u (h) = Fh ∘ u

- **characteristic functions** χ: X → { 0, 1 }
Characteristic functions are 1-1 with subsets.
Hence, can be represented by the (contravariant) **powerset**

	Powerset functor	
	P^op: Set^op → Set	
object	P^opX = 2^X	
arrow	P^op f: 2^X → 2^X'	any f: X' → X
	P^op f(S) = f^-1 (S) for S ⊆ X	



Representation:

φ: { _ → { 0, 1} }^op ⇒ P^op
φX: { X→ {0,1} }^op → 2^X for any set S
φX(χS) = -χS {1} = S ⊆ X χS and S ⊆ X have 1-1 relation
χS(x) = 1 iff x ∈ S

base map
φ = P^op * J where J = 2^{0,1} = { {0}, {1}, {0,1} }

naturality
φ (χS ∘ f) = P^op f ($\varphi\chi$S) for any S ⊆ X
= S' ⊆ X' f: X' → X, S' = f^-1(S)

- **cyclic functions (x)** + **forgetful functor U: A → Set**
Cyclic functions generated by a single element $x \in A$

(x): Z → A
(x) (n) = nx

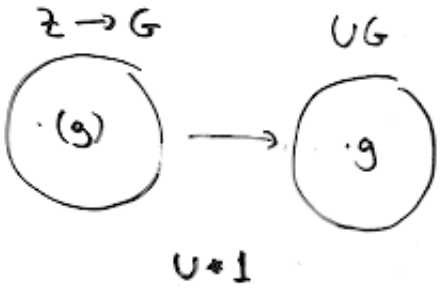
can be represented by the single element $x \in A$
i.e. by the **forgetful functor U: A → Set**

e.g. **U: Grp → Set**
For any group G and $g \in G$, there is exactly **one cyclic homomorphism**

(g): Z → G
(g) (1) = g

Hence a representation is

U * 1: { Z → _ } ⇒ U
U * 1: { Z → G} → UG
U * 1 (g) = U(g) (1) = g for any $g \in G$



e.g. **U: Vect → Set** for Vect over **field K**
Similarly, if (v): K → V for any $v \in V$ is the generator, then

U * 1: { K → _ } ⇒ U
U * 1: { K → V } ⇒ UV
(U * 1) (v) = U(v) (1) = v for any $v \in V$

- **group action** (aka **G-set**)
Ref: https://en.wikipedia.org/wiki/Representable_functor

Let $\cdot$: G x X → X be a group action. Then α can be viewed in 2 forms:

$g \cdot x \in X$

Action by g	**Action at x**
g*: X → X	*x: G → X
g*(x) = gx	*x(g) = gx

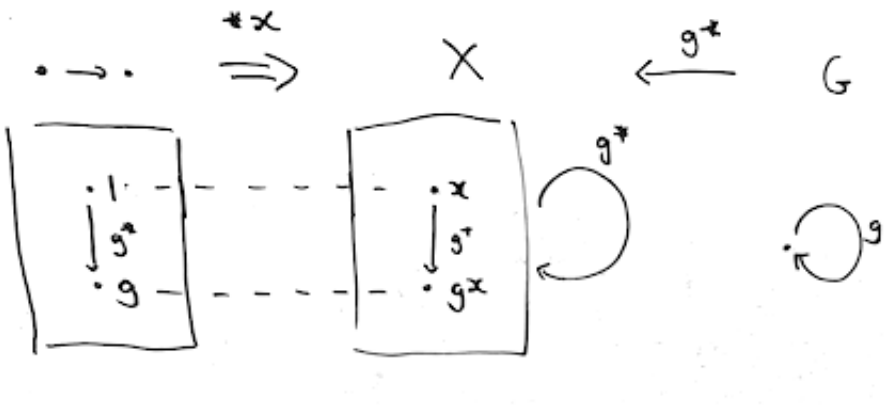
In categorical form:

Group G
G = { ● } single object ●
g: ● → ● arrow for each $g \in G$

Action functor
A: G → Set
A• = X
Ag: X → X
Ag = g*

Action at x transform
*x: { • → _ } ⇒ A for any x ∈ X
*x: { • → • } → X since G only one object •
x(g) = gx

Note: every transformation φ: { • → _ } ⇒ A is of the form φ = *x by Yoneda



Then a dual ***x: { • → _ } ⇒ A** is a representation of G if it is isomorphic.
i.e. if there is a 1-1 relation

g ∈ G <=> y ∈ X
gx = y

By group theory:

Proposition: group representation
*x is a **representation** iff the group action is free and transitive.

Note:

free for any g, x
gx = x <=> g = 1

transitive for any x, x' ∈ X, there exists g
x' = gx

==========
proof
Idea:
free ~ *x is injective
transitive ~*x is surjective

<=) Suppose G is simple and transitive
Then

- *x is injective:
Suppose *x(g) = *x(g') for some g, g'.
Then

gx = g'x
=> g'^-1gx = x
=> g'^-1g = 1 since simple
i.e. g = g'

- *x is surjective:
For any y ∈ X, by transitivity, there exists g such that

*x(*g) = gx = y

Hence *x is a representation

=> Suppose *x is a representation
- free
Suppose gy = y for any g, y
Let y = g'x = *x(*g) for by *x representation. Then

g(g'x) = (g'x)
=> g'^-1gg'x = x
=> g'^-1gg' = 1 by representation
=> gg' = g'
=> g = 1

- transitive
Suppose any y, y' ∈ X
Let y = gx = *x(*g), y' = g'x = *x(*g) by *x representation
Then.

(g'g^-1) y = g'g^-1gx
= g'x
= y'

QED
==========

==========

----- Natural isomorphism aka adjoint-----

Adjoint
Adjoint is just a **natural** family of isomorphic base maps
i.e.
Let
F, A → B, G: B → A functors
ηX: X → GFX

The **adjoint <F, G, η>** is the family of base maps

<F, G, η> = { G * ηX ∈ X → G[FX→_] | X ∈ A }

where

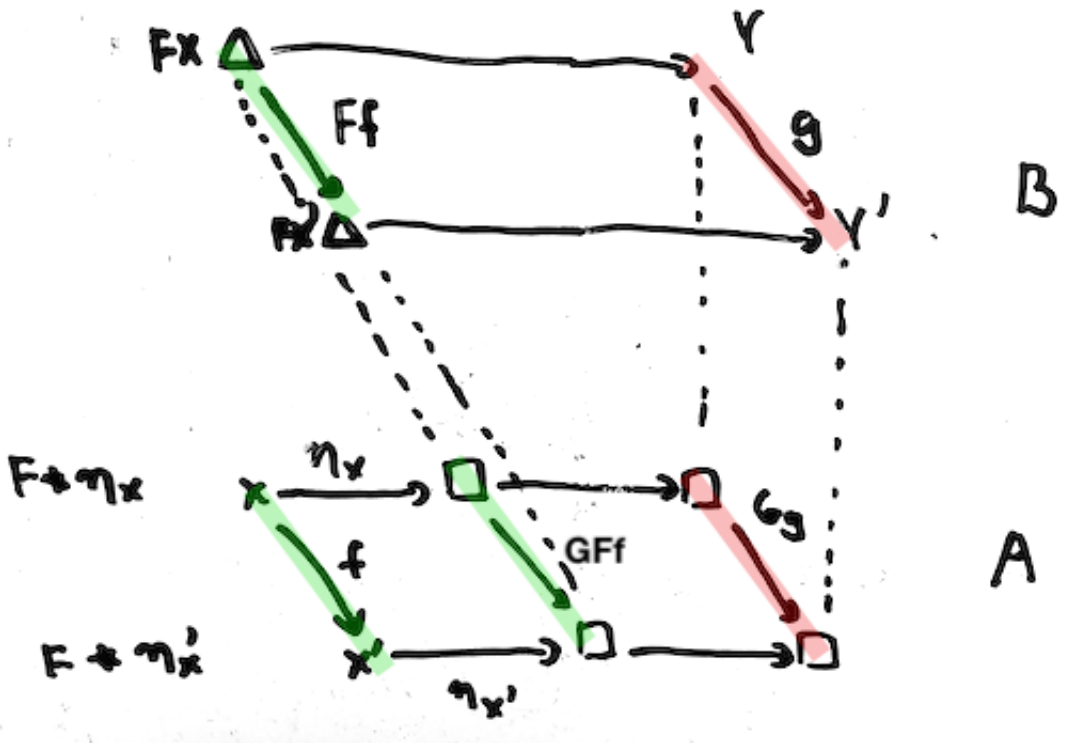
naturality G * ηX is natural in X
i.e. if f: X → X', then

***f ∘ (G * ηX') = (G * ηX) ∘ *Ff** base map form

η**X'** ∘ **f = GFf** ∘ η**X** base form

isomorphism G * ηX is isomorphic for each X ∈ A

natural in X and Y



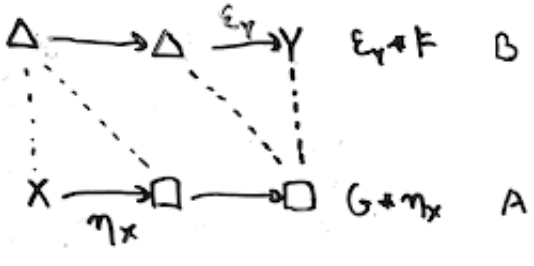
Theorem: inverse
If <F, G, η> is an adjoint, then each base has an **inverse co-base**

-(G * ηX) = ε**Y * F** ∈ F[_→GY]→Y

where

counit εY: FGY → Y
εY = -(G * ηGY) (1GY)

naturality εY is natural in Y
i.e. g ∘ εY = εY' ∘ FGg for g:Y→Y'



==========
Proof
- **co-base map**

G * ηX is isomorphic
=> G * ηX is a representation i.e. isomorphic + natural in Y
=> -(G*ηX) is a representation
=> -(G*ηX) is an isomorphic co-base with base εY = - (G*ηX) (1GY): FGY → Y

- εY is natural in Y
Just by **naturality** of G*ηX

Gg* ∘ (G*ηX) = (G*ηX) ∘ g* naturality

=> invert
-(G*ηX) ∘ Gg* = g* ∘ -(G*ηX)
i.e. (εY*F) ∘ Gg* = g* ∘ (εY*F)

=> εY ∘ FGg = (εY*F) ∘ Gg (1GY)
= g* ∘ (εY*F) (1GY)
= g ∘ εY

QED
==========

Theorem: uniqueness upto isomorphism

<F, G, η>, <F', G, η'> adjoints => F ≅ F'
<F, G, η>, <F, G', η'> adjoints => G ≅ G'

==========
Proof
Just by uniqueness of isomorphic base maps
For any X

G*ηX: {FX→_} ⇒ {X→G_} isomorphic base maps
G*η'X: {F'X→_} ⇒ {X→G_}

=> isomorphism by uniqueness of isomorphic base maps

σX: FX→FX'
σX = -(G*ηX) (η'X)
= (εY * F) (η'X)

and σX is natural in X since it is a value of natural co-base εY * F: {_→GY} ⇒ {F_→Y}

=> σ: F ⇒ F' isomorphic

QED
==========

Theorem: identities
Let
h: FX → Y
h': X → GY

- **Naturality**

For f: X'→X g:Y→Y'

base form

ηX' ∘ f	=	GFf ∘ ηX	
g ∘ εY	=	εY' ∘ FGg	
G * (ηX' ∘ f)	=	G * (GFf ∘ ηX)	with base maps
(g ∘ εY) * F	=	(εY ∘ FGg) * F	

base map form

*f ∘ (G * ηX')	=	(G * ηX) ∘ Ff*	
g* ∘ (εY * F)	=	(εY * F') ∘ *Gf	

applied form

Gh ∘ (ηX' ∘ f)	=	G(Ff ∘ h) ∘ ηX	
(g ∘ εY) ∘ Fh'	=	εY' ∘ F(h' ∘ Gf)	

- **Identity**: applied form

h	=	εY * F (G * ηX (h))
	=	εY ∘ FGh ∘ FηX
h'	=	G * ηX (εY * F(h'))
	=	GεY ∘ GFh' ∘ ηX
1GY	=	GεY ∘ ηGY
1FX	=	εFX ∘ FηX
1GY	=	G * ηGY (εY)
1FX	=	εY * F (ηX)

- **Identity**: transformation form

Let

ε: FG ⇒ IB	η: IA ⇒ GF
ε * F: F ⇒ IB	G * η: IA ⇒ G

Base map

IA	=	(ε * F) ∘ (G * η)
IB	=	(G * η) ∘ (ε * F)

Base

1G	=	Gε ∘ ηG
1F	=	εF ∘ Fη

Lemma: Extension

Let

G: B → A	functor
ηX: X → GFX where FX ∈ B	

If G * ηX is an isomorphic base map for each X ∈ A, then

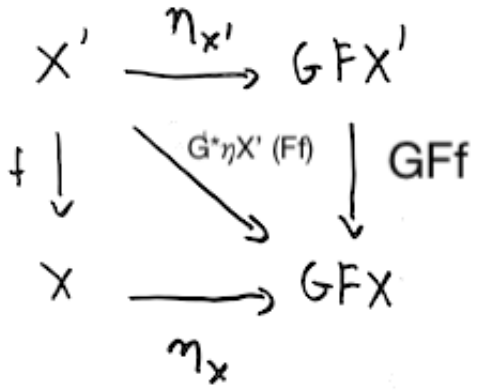
ηX is **natural in X**	<=>	**Ff: FX' → FX**	for any f: X'→X

Ff = -(G * ηX') (ηX ∘ f)

i.e. G * ηX (Ff) = ηX ∘ f

Hence, there is a **unique extension** of F to a functor A → B such that <F, G, η> is an adjoint

ηX is **natural in X** <=> there exists unique F: A → B such that <F, G, η> adjoint



==========

Proof

- naturality

GFf ∘ ηX' = G * ηX' (Ff)

=>

natural in X <=> G * ηX' (Ff) = ηX ∘ f

GFf ∘ ηX' = ηX ∘ f

- functor F: A → B

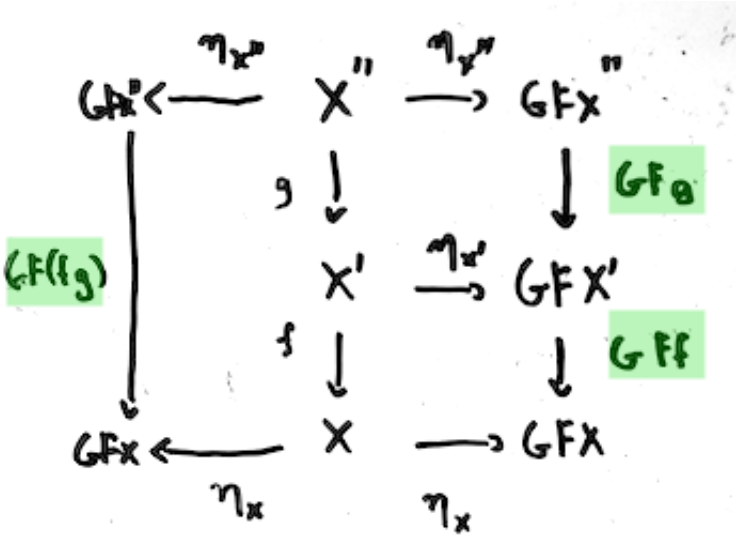
claim: F1X = 1FXX

Since G * ηX (1FX) = ηX

claim: F preserves composition

G * ηX'' (F(f ∘ g))	=	ηX ∘ f ∘ g	
G * ηX'' (Ff ∘ Fg)	=	GFf ∘ GFg ∘ ηX''	
	=	GFf ∘ (ηX' ∘ g)	naturality
	=	(ηX ∘ f) ∘ g	naturality

=> F(f ∘ g) = Ff ∘ Fg since G * ηX'' is 1-1



QED

==========

Theorem: characterization

Let

G: B → A functor

ηX: X → GFX where FX ∈ B

a) **adjoint** <F, G, η>: A → B adjoint

with inverse εY: FGY → Y

b) **extension** G * ηX ∈ X→G[FX→_] isomorphic for each X ∈ A

where

G:B→A functor

F:A→B induced functor by extension

c) **co-extension** εY * F ∈ F[_→GY]→Y isomorphic for each Y ∈ B

where

F: A→B functor

G: B→A induced functor by extension

d) **Identity** 1F = εF ∘ Fη

1G = Gε ∘ ηG

where

F: A→B G:B→A functors

η: IA ⇒ GF ε: FG ⇒ IB

==========

Proof

- Identity => adjoint

Suppose identity.

claim: G * ηX is isomorphic

****

Idea: Just use the inverse εY * F

- injective

Suppose any h: FX → Y

$(\varepsilon Y * F)\ (G * \eta X\ (h))$	=	$\varepsilon Y \circ FGh \circ F\eta X$	
	=	$(h \circ \varepsilon FX) \circ F\eta X$	by naturality
	=	h	identity

=> injective

- surjective
Suppose any h: X → GY. Let

$h' = \varepsilon Y * F\ (h)$: FX → Y

=>

$G * \eta X\ (h')$	=	$G(\varepsilon Y \circ Fh) \circ \eta X$	
	=	$G\varepsilon Y \circ (\eta GY \circ h)$	by naturality
	=	h	identity

QED claim
****

Also $G * \eta X$ is natural in X since η is natural
=> { $G * \eta X$ } is an adjoint

QED
==========

Proposition: full/faithful
Let η: I → GF, ε: FG → I

F is **faithful** <=> η is **monic**

F is **full** <=> ε**FX** has a preimage
i.e. there exists eX: GFX → X such that ε**FX = FeX** for each X

<=> η is **split epic**
$\eta X \circ eX = 1GFX$

G is **faithful** <=> ε is **epic**

G is **full** <=> η**GY** has a preimage
i.e. there exist nY: Y → FGY such that η**GY = GnY** for any Y

<=> ε is **split monic**
$nY \circ \varepsilon Y = 1FGY$

Hence

F is **full** and **faithful** <=> η **isomorphic**

G is **full** and **faithful** <=> ε **isomorphic**

==========

Proof:

- **faithful +** η

$\eta X \circ f = GFf \circ \eta X'$ by naturality

$= G * \eta X' (Ff)$

Hence

η is monic <=> $G * \eta X' (Ff') = G * \eta X' (Ff')$ implies f =f '

<=> Ff = Ff' implies f = f' since G * ηX' is isomrphic

i.e. faithful

- **full** + η

Let h: FX → FX'. Use the adjoint to **map** it into F:

h = -$\varphi\varphi$h

= εFX' ∘ Fφh

= F(eX' ∘ φh) if εFX' = FeX for some eX': GFX' → X'

=>

F is **full** <=> εFX = FeX for some eX: GFX → X

<=> 1GFX = η**X** ∘ **eX** by **naturality** ηX ∘ eX = GεFX ∘ ηGFX = 1GFX

i.e. ηX is split epic

- isomorphism

By general property, **monic + split epic** implies isomorphic

i.e.

$\eta X = (\eta X \circ eX) \circ \eta X$

$= \eta X \circ (eX \circ \eta X)$

$\eta X = \eta X \circ 1$

=> (eX ∘ ηX) = 1 since ηX is monic

QED

==========

Corollary: invertible

If G is full, then ηG and Gε are inverses i.e.

Gε ∘ ηG = 1G ηG ∘ Gε = 1GFG

If F is full, then Fη and εF are inverses i.e.

εF ∘ Fη = 1F Fη ∘ εF = 1FGF

==========

Proof:

Idea: use natural + identity

- G is full

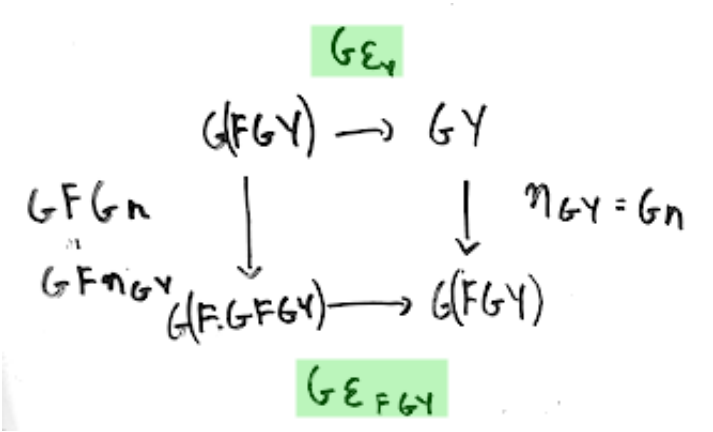
Let ηGY = GnY for some n: Y → FGY. Then

ηGY ∘ GεY = G(nY ∘ εY)

= G(εFGY ∘ FGnY) naturality of εY

= G(εFGY ∘ FηGY)

	=	1GFGY	identity



- F is full
Let $\varepsilon FX = FeX$ for some $e: GFX \to X$. Then

$F\eta X \circ \varepsilon FX$	=	$F(\eta X \circ eX)$	
	=	$F(GFeX \circ \eta GFX)$	naturality of ηX
	=	$F(G\varepsilon FX \circ \eta GFX)$	
	=	1FGFX	identity

QED
==========

Examples:
- **Forgetful functor U: A → B**
More (A) to less (B) structure

faithful since Uf: UX → UX', f: X → X' have the **same value**
NOT full usually no

e.g.
interpretation of free vector W → Free(UW) is surjective

- **Inclusion functor** I: S → A for S ⊆ A
Like forgetful, more (S) to less (A) structure

faithful since If: IX → IX' and f: X → X' are exactly the same (both value and structure)
full if S is a full subcategory

e.g.
abelianization of A ∈ Ab q: A/[A,A] → A is isomorphic

floor of integer n ∈ Z ⌊n⌋ <= n is isomorphic
i.e. ⌊n⌋ == n

ceil of integer n ∈ Z n <= ⌈n⌉ is isomorphic
i.e. n == ⌈n⌉

-- **Diagonal functor Δ: J → A**
Faithful
Full if connected

e.g.

split of product	δ: X → X x X is injective
join of co-product	δ: X⊔X → X is surjective

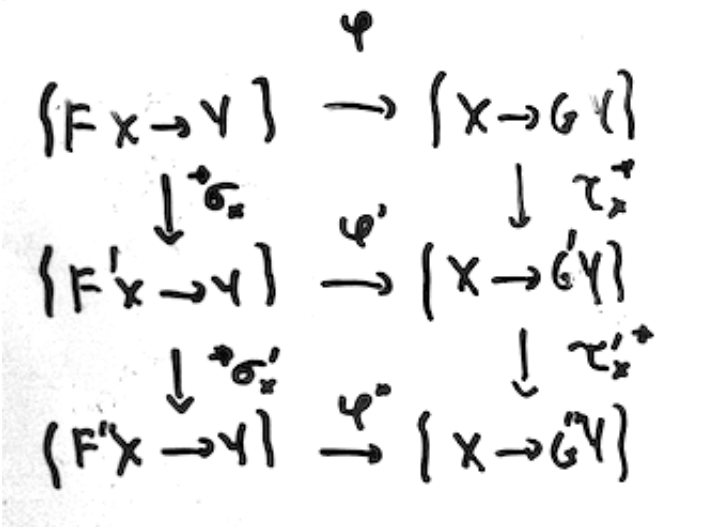
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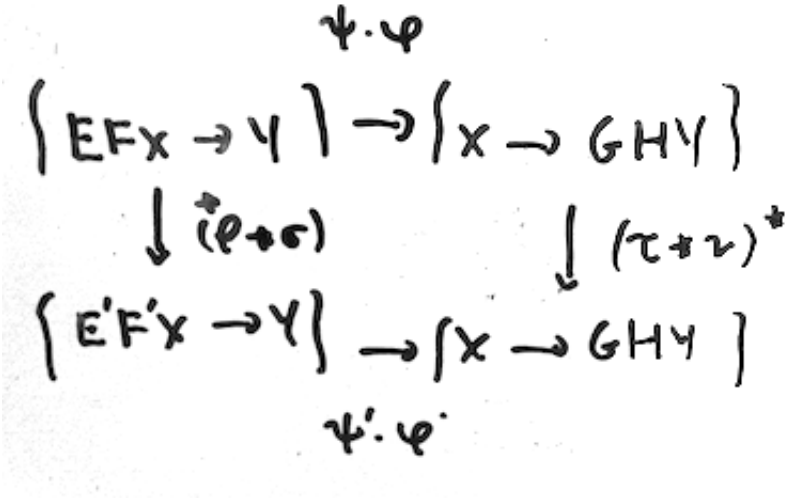
=== Adjoint categories ===

*** Adjoint categories**

G, G': B → A H, H': C → B
τ: G ⇒ G' υ: H ⇒ H'

	Cat	**Adj**
objects	A ∈ Cat	A ∈ Cat
morphisms	G: B → A	<F, G, η>: B → A
compositions	GH: C → B → A	<F, G, η> ∘ <E, H, η'> : C → A <F, G, η> ∘ <E, H, η'> = <EF, GH, GηF ∘ η'>
	A^B i.e. Cat(B, A)	**Adj(B, A)**
objects	G: B → A	<F, G, η>: B → A
morphisms	τ: G ⇒ G' where τY' ∘ Gg = G'g ∘ τY	(σ, τ): <F, G, η> ⇒ <F', G', η'> where τ* ∘ φ = φ' ∘ *σ
composition	τ' ∘ τ: G ⇒ G	(σ', τ') ∘ (σ, τ): <F, G, η> ⇒ <F'', G'', η''> (σ', τ') ∘ (σ, τ) = (σ ∘ σ', τ' ∘ τ)
vertical composition	τ' ∘ τ: G ⇒ G' ⇒ G'' (τ' ∘ τ)Y = τ'Y ∘ τY	(σ', τ') ∘ (σ, τ): <F, G, η> ⇒ <F', G', η'> ⇒ <F'', G'', η''> (σ', τ') ∘ (σ, τ) = (σ ∘ σ', τ' ∘ τ)
horizontal composition	τ ∗ υ: GH ⇒ G'H' τ ∗ υ = τH' ∘ Gυ = G'υ ∘ τH	(σ, τ) ∗ (ρ, υ): <EF, GH, GηF ∘ η'> ⇒ <E'F', G'H', GρF ∘ ρ'> (σ, τ) ∗ (ρ, υ) = (ρ ∗ σ, τ ∗ υ)





==========

===[TO UPDATE] Examples ===

Examples

- Product XxY is a universal property

XxY ∈ A	universal object - product
(p, q): (XxY, XxY) → (X, Y)	universal arrow - projections
(p, q) * Δ: {_ → XxY} → {Δ_ → (X, Y)}	universal property
(p, q) * **Δ** f: **Δ**Z → (X, Y)	for f: Z → XxY
(p, q) * **Δ** f = (p, q) ∘ **Δ**f	
= (pf, qf)	

- Coproduct X⊔Y

X⊔Y
(i, j): (X, Y) → (X⊔Y, X⊔Y)

Δ * (i, j): { X⊔Y → _ } → { (X, Y) → **Δ**_ }
Δf * (i, j): (X, Y) → **Δ**Z
Δ * (i, j) f = **Δ**f ∘ (i, j)
= (f ∘ i, f * j)

- Free vector space VX is a universal property

VX ∈ Vect — universal object - free vector space
u: X → VX — universal arrow

U * u: {VX → _} → {X → U_} — universal property
U * uf: X → UW — for f: VX → W
U * u f(x) = f(x)

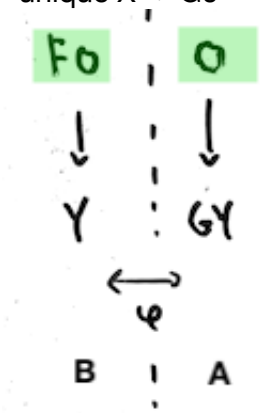
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=== [TO UPDATE] Adjoints and limits ===

Corollary

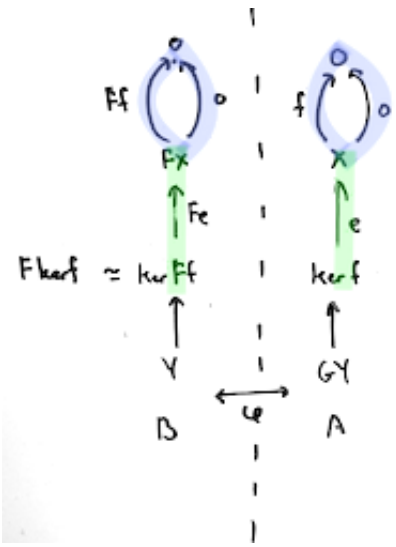
initial/terminal
F0 is **initial** — if 0 is initial
unique F0 → Y fo

G0 is **terminal** — if 0 is terminal
unique X → G0



monic/epic
Ff: FX → FX' is **epic** — if f: X → X' is epic
Gg: GY → GY' is **monic** — if g: Y → Y' is monic

kernel
ker Ff ≅ F(ker f)
Coker Gg ≅ G(Coker g)



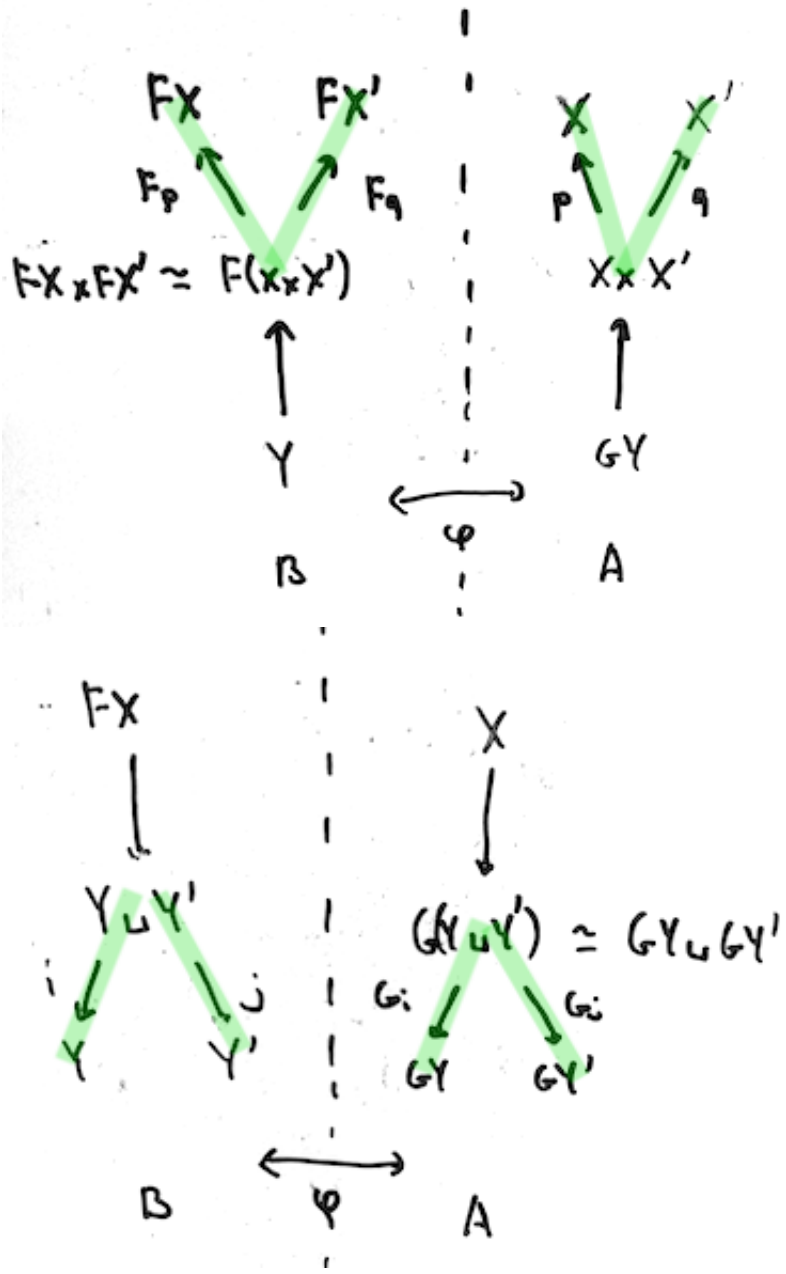
product/coproduct
FX x FX' ≅ **F (X x X')** — if X x X' ∈ A exists

projections
  Fp: FX x FX → FX
  Fq: FX x FX' → FX'

projections (p,q): X x X' → (X, X')

GY ⊔ GY' ≅ G (Y ⊔ Y')
injections
  Gi: GY → GY ⊔ GY'
  Gj: GY → GY ⊔ GY'

if Y ⊔ Y' ∈ B exists
  injections (i, j): (Y, Y') → Y ⊔ Y'



==========

=== OLD ===

****

==========

This
is
code
Monaco 14

Proposition: representation

as element **x** ∈ **FX**	<=>	as transformation **F * x: {X→_} ⇒ F**
element x ∈ FX		**object** F * x: {X→_} ⇒ F
arrow Ff: FX0 → FX for f: X0 → X		**arrow** *f: {X→_} → {X0→_} for f: X0 → X
1-1 x = Ff(x0)		**1-1** (F * x0) ∘ *f = F * (f(x0)) = F * x i.e. x = Ff(x0)

Hence

[**representation**]		[**universal**]
φ: {X0 →_} ⇒ F is a representation	<=>	x0 ∈ FX0 is a universal element
i.e. bijection	φ = F * x0	
	<=>	[**Yoneda embedding**] F * x0: {X0→_} ⇒ F is a terminal

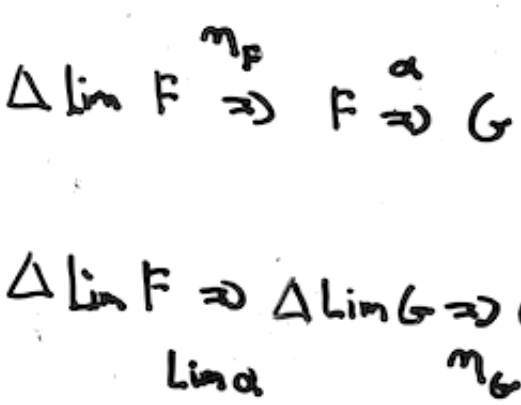
element in Hom ↓ 1F

Proposition: naturality
Limits are natural, if they exist.
i.e. if Lim F exists for all F ∈ B^A, then

Limit functor
Lim: A^J → J
Lim α: Lim F → Lim G α: F ⇒ G

Colimit functor
Colim: A^J → J
Colim α: Colim F → Colim G α: F ⇒ G

naturality
Lim (β ・ α) = Lim β ∘ Lim α for any α: F ⇒ G
Colim (β ・ α) = Colim β ∘ Colim α β: G ⇒ H



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Proof

- **Lim α definition**

Let α: F ⇒ G. Then

$\alpha \cdot \eta$F: **Δ**Lim F ⇒ G

=> there is a unique σ: Lim F → Lim G

$\alpha \cdot \eta$F	=	ηG · **Δ**σ	
	=	ηG · **Δ**Lim α	define Lim α := σ

- **naturality**

Let α: F ⇒ G, β: G ⇒ H

($\beta\alpha$) ∘ ηF: **Δ**Lim F ⇒ H

=>

($\beta\alpha$) ηF	=	ηH **Δ**Lim $\beta\alpha$	by definition of Lim H
($\beta\alpha$) ηF	=	β (ηG **Δ**Lim α)	by definition of Lim G on (α ∘ ηF): **Δ**Lim F ⇒ G
	=	(β ηG) **Δ**Lim α	by definition of Lim H on (β ∘ ηG): **Δ**Lim G ⇒ H
	=	ηH **Δ**Lim β **Δ**Lim α	

=> Lim $\beta\alpha$ = Lim β Lim α by uniqueness

QED

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Composition

(F' * u') ∘ (F * u)	=	F'F * (F'u ∘ u')	Conj Y'→F'F[X→_] where F'Y = F'X'
	=	*u' ∘ (F'F * F'u)	

F'u ∘ u': Y'→F'FX

(w' * G') ∘ (w * G) = (w' ∘ G'w) * G'G Conj G'G[_→V]→W' where G'W = G'V'
= w'* ∘ (G'w * G'G)
w' ∘ G'w: G'GV→W'

(w * G) ∘ (F * u) = *Gu ∘ (w * GF) {X→Z} → {GY→W} where FZ = V
= w* ∘ (GF * Gu)
(w * G) ∘ (F * u) (h) = w ∘ GFh ∘ Gu

(F * u) ∘ (w * G) = Fw* ∘ (FG * u) {Z→V} → {Y→FV} where GZ = X
= *u ∘ (Fw * FG)
(F * u) ∘ (w * G) (h) = Fw ∘ FGh ∘ u

Special cases
*g ∘ (F * u) = F * (u ∘ g) change Y→_ to Y'→_ with g: Y' → Y
(F * u) ∘ *f = F * (Ff ∘ u) change X→_ to X'→_ with f: X → X'
*f ∘ *g = *(f ∘ g)

(w * G) ∘ g* = (w ∘ Gg) * G change _→V to _→V' with g: V'→V
f* ∘ (w * G) = (f ∘ w) * G change _→W to _→W' with f: W→W'
f* ∘ g* = (f ∘ g)*

(F * u) ∘ f* = Ff* ∘ (F * u) change X→_ to X→Z with h: X → Z, f: Z → Z'
= (Ff * F) ∘ *u Y→F_ to Y→FZ'
(F * u) ∘ f* (h) = Ffh ∘ u
g ∘ f = f* ∘ *g

(w * G) ∘ *g = *Gg ∘ (w * G) change _→V to Z→V with h:Z→V, g: Z'→Z
= w* ∘ (G * Gg) G_→W to GZ'→W
(w * G) ∘ *g (h) = w ∘ Ghg
f* ∘ *g = *g ∘ f*

- natural in X→X'
If
G: B → A X = GY
G': B →A X' =G'Y
τ: G' ⇒ G

then

(F' * u') ∘ τX* = F' * (F'τX ∘ u')
= F' * (s ∘ u)

Hence

σ* ∘ (F * u) = (F' * u') ∘ τX*

Horizontal transformation: natural σ: F ⇒ F'

Let

F:A→B F':A→B
F * u ∈ Conj Y→F[X→_]
F' *u' ∈ Conj Y→F'[X'→_]

If there exist transformations (**t:X'→X,** σ**: F ⇒ F')** where

$\sigma^* \circ$ **(F * u) = (F' ∘ u') ∘ *t**

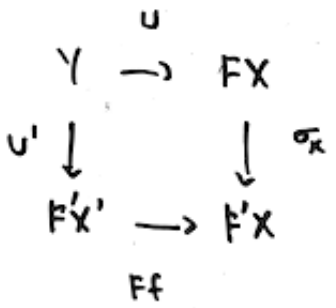
then (t, σ) is a **horizontal transformation** of F*u to F*u'

Theorem: units
Horizontal transformation is determined by its **units**.
i.e. If (t:X'→X, σ: F ⇒ F') then

(t, σ) is a horizontal transformation <=> $\sigma X \circ u = Ft \circ u'$

Hence

F' * (σX ∘ u) = F' * (Ft ∘ u') ∈ Conj Y→F'[X→_]



==========
proof
<=) Suppose $\sigma X \circ u = Ff \circ u'$
Then for h: X→Z

$\sigma Z \circ (F * u)\ (h)$	=	$\sigma Z \circ Fh \circ u$	
	=	$F'h \circ \sigma X \circ u$	naturality of σ
	=	$F'h \circ Ff \circ u'$	
	=	$(F^* u') \circ {}^*f\ (h)$	

=>) Suppose (f, σ) horizontal
$\sigma X \circ u = \sigma X \circ (F^*u)\ (1) = (F'^*u') \circ {}^*f\ (1) = Ff \circ u'$

QED
==========

Proposition: Horizontal transformation
If
G: B → A X = GY
G': B →A X' =G'Y
τ: G' ⇒ G

then

$\sigma^* \circ (F * u) = (F' * u') \circ \tau^*$

OLD

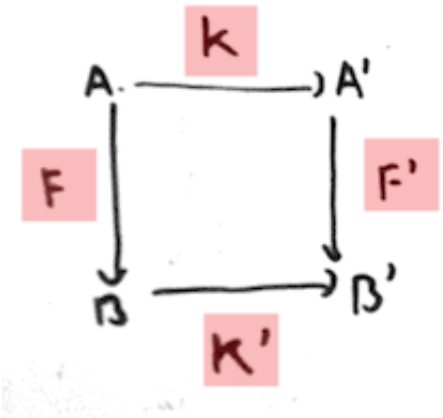
Vertical transformation: A → B → C
Let
F:A→B F * u ∈ Conj Y→F[X→_]
F':A'→B' F' * u' ∈ Conj Y'→F'[X'→_]

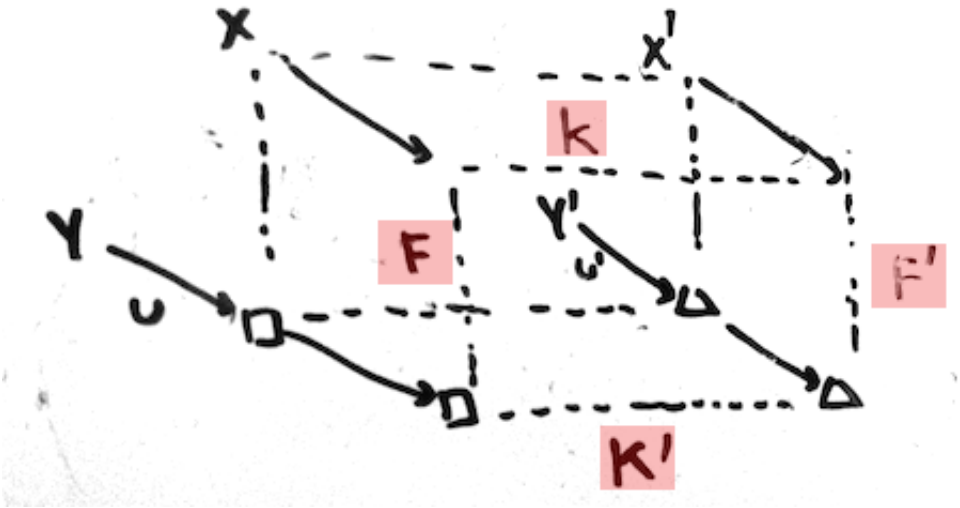
If the pair of **functors (K':A→A' K:B→B')** satisfy

KF = F'K'

K* ∘ (F * u) = (F' * u') ∘ K'*
hence X' = K'X
Y' = KY

then (K', K) is a **vertical transformation** of F*u to F'*u'





Yoneda Lemma
Yoneda Lemma just states that every **natural transformation** φ: {X→_} ⇒ {Y→F_} for F: A → B is a **conjugate**

Proposition: natural

Composition

Offset
For u: Y → FX0

F * u: {X0→_} → {Y→F_}
F * u (f) = Ff ∘ u

[OLD]

Let F: A → B
Conjugate of A to B at **base (arrow)** u: Y → FX is

F * u: {X→_} → {Y→F_}
F * u (f) = Ff ∘ u for f: X→X'

F * u(f) is called the **conjugate** of f.

Co-conjugate of A to B at **base** u: FX → Y

u * F: {_→X} → {F_→Y}
u * F(f) = u ∘ Ff for f: X→X'

u * F(f) is the **co-conjugate** of f.

Conjugate
Conjugate base element
Let F: A → Set
Conjugate of A at **base element** x ∈ FX is

F * x: {X→_} → F_
F * x (f) = Ff(x) for f: X→X

Proposition: basic

Identity
F * u (1) = u

Composition (naturality)
F * u (h ∘ g) = Fh ∘ (F * u (g))

F * (u ∘ f) = (F * u) ∘ *f natural in Y' → Y
where f: Y' → Y

case: If u: I ⇒ FHY is natural

F * (FHh ∘ u) = (F * u) ∘ *h natural in FHY' → FHY

G * (σX ∘ u) = σ ∘ (F * u) natural in F ⇒ G

where σ: F ⇒ G

Conjugate functor

*: { u: Y → FX | any Y, F } → { conjugates }

* u = F * u object

* (σ: F ⇒ G, f: Y → Y') : F * u → G * u' arrow

where u' = G * (σX ∘ u ∘ f)

Proposition: Category <-> Set correspondence

Let

F: A → B

G: A → Set

GX = {Y→FY}

Equivalent

universal arrow	**universal element**	**natural transformation**
u: Y→FX	x0 = u	φ: {X→_} ⇒ {Y→F_}
F * u: {X→_} → {Y→F_}	G * x: {X→_} ⇒ G	φ = F * u
		u = φ(1X)

==========

---- NOTES ----

universal property

B A

h: Y → FX <=> f: GY → X

h = Ff ∘ ηY

unit ηY: Y → FGY

naturality in Y

ηY' ∘ g = FGg ∘ ηY for any g: Y → Y'

Yoneda

φ: {X0→} ⇒ F <=> xφ ∈ FX0

φ1X0 = xφ

φf = Ff(xφ)

****

****

universal property
Let F: A → B
u: Y → FY0 such that F + u is bijective

F * u: { Y0 → _ } → { Y → F_ }
F * u (f) = Ff ∘ u

Natural in Y if:

u' ∘ g = Fg ∘ u for any g: Y → Y'

Yoneda
Let F: A → Set
Every φ: {X0→} ⇒ F is of the form

φ **= F * x0**: {X0→} ⇒ F x0 = φ1X0
F * x0 (f) = Ff (x0)

{ φ } is natural in X0 and F
For any f: X0 → X0'

***f: { F * x0 } → { F * x0' }**
*f (F * x0) = F * (f(x0))

For any σ: F ⇒ G

σ***: { F * x0 } → { G * x0 }**
σ*(F * x0) = G * (σ(x0))

Adjoint
F: A → B, G: B → A

φ: {FX →Y} → {X→GY}
φ = G * ηX ηX: X → GFX
-φ = εY * F εY: FGY → Y

Limit
F: A → B
Universal cone u: **Δ**Lim F ⇒ F

u * **Δ**: {_ → Lim F} → {**Δ**_ ⇒ F}